

TAKEOFF

- ☐ Fuel Pump — ON
- ☐ Fuel Tank - FULLEST TANK
- ☐ Lights — AS REQUIRED
- ☐ Flaps — SET 10° (25° SHORT/SHORT)
- ☐ Fuel/Oil Gauges - GREEN
- ☐ Airspeed Alive — CHECK
- ☐ Rotate — **65-70 MPH**
- ☐ Climb — **Vy 105 MPH / Vx 95 MPH**

IFR — DEPARTURE / ENROUTE

- ☐ Clearance — REVIEW (route / altitude / frequencies)
- ☐ Bugs — HDG / ALT / MINs SET
- ☐ Navigation Source — CONFIRM
- ☐ Instruments — CHECK (gyro / CDI sensing)
- ☐ Fuel Pump — ON
- ☐ Lights — AS REQUIRED

IFR — APPROACH CONFIGURATION

- ☐ Power — SET
- ☐ Pitch — SET
- ☐ Configuration — FLAPS 10°
- ☐ Airspeed — 110 MPH
- ☐ Sink Rate — 500 FPM
- ☐ Garmin 530W approach activation — LOAD / VERIFY
- ☐ DADS — CHECK every 1 NM or 500 FT
- ☐ 6 Ts — APPLY / REVIEW on change

Goal: Stabilized by FAF

IFR PERFORMANCE

Phase	Power	Pitch	Config	Speed	V/S
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PREFLIGHT

FLOW

- ☐ Covers & Wheel Chalks - REMOVE
- ☐ Paperwork - ARROW-AV1ATE
- ☐ Control Lock - REMOVE
- ☐ Flaps - DOWN
- ☐ Trim - TAKEOFF
- ☐ Master - ON
- ☐ Lights - ON & CHECK
- ☐ Pitot Heat - ON & CHECK
- ☐ Stall Light - CHECK
- ☐ Fuel Quantity — CHECK / PROPER TANK
- ☐ Lights - OFF (KEEP BEACON ON)
- ☐ Pitot Heat - OFF
- ☐ Master - OFF
- ☐ Airframe & Hinges - WALK AROUND & CHECK
- ☐ Cowling - SECURE
- ☐ Tires & Brakes - CHECK
- ☐ Prop and Intake - CHECK
- ☐ Fuel Tanks - SUMP, MEASURE
- ☐ Oil — CHECK (6-10 QTS)
- ☐ Tow Bar - STOW
- ☐ Baggage - SECURE

SAFETY CHECKLIST

- ☐ Seatbelts - EXPLAIN & ON
- ☐ Air vents/Heat - LOCATE & EXPLAIN
- ☐ First Aid & Fire Extinguisher - LOCATE
- ☐ Exits - EXPLAIN & LOCK
- ☐ Talking - ASSIST TRAFFIC
- ☐ Your Questions - ASK

ENGINE START

- ☐ Brakes - SET
- ☐ Flaps - UP
- ☐ Electrical - ALL OFF
- ☐ Circuit Breaker - ALL IN
- ☐ Carburetor Heat - OFF
- ☐ Controls — FREE / CORRECT
- ☐ Prop - FULL FORWARD
- ☐ Mixture - RICH
- ☐ Fuel Selector - STARTUP SIDE
- ☐ Master - ON
- ☐ Throttle - 1/4 INCH OPEN
- ☐ Key/Mags - BOTH
- ☐ Area - CLEAR PROP
- ☐ Fuel Pump - ON
- ☐ Primer - 2 TO 3 OR AS REQUIRED
- ☐ Starter - CRANK ENGINE
- ☐ Throttle - 1000 RPM
- ☐ Avionics Master - ON
- ☐ Fuel Pump - OFF

PROFILES

GROUND CHECK

- ☐ Doors & Windows - CLOSED
- ☐ Seatbelts - SECURE
- ☐ Controls - CHECK
- ☐ Trim - TAKEOFF
- ☐ Airspeed Indicator - READS 0
- ☐ Attitude Indicator - <5° BANK AND WINGS LEVEL
- ☐ Altimeter - FIELD ELEVATION
- ☐ VSI - Zero
- ☐ Turn & Slip Indicator - FLAGS OFF, WINGS LEVEL, FULL FLUID
- ☐ Heading Indicator - SET TO COMPASS HEADING
- ☐ Garmin 530W database — CHECK / UPDATE
- ☐ Garmin 530W instrument check — VERIFY

RUN-UP

- ☐ Brake — SET
- ☐ Throttle — 2000 RPM
- ☐ Magnetos — CHECK (175 RPM MAX/50 RPM DIFFERENCE)
- ☐ Propeller — CHECK / FULL FORWARD
- ☐ Fuel/Oil Gauges — GREEN
- ☐ Ammeter - POSITIVE AMPS
- ☐ Suction — CHECK 5.0" +/- 1" HG
- ☐ Carburetor Heat — ON
- ☐ Throttle — IDLE / SMOOTH
- ☐ Carburetor Heat - OFF
- ☐ Throttle - 800-1000 RPM
- ☐ Flight Plan - ACTIVATE
- ☐ Stable RPM
- ☐ No abnormal vibration
- ☐ Engine indications normal

ENGINE FAILURE — IN FLIGHT

- ☐ Radio call — MAYDAY, MAYDAY, MAYDAY on 121.5
- ☐ Squawk — 7700
- ☐ Airspeed **100 MPH (BEST GLIDE)**
- ☐ Fuel Selector — FULLEST TANK
- ☐ Fuel Pump — ON
- ☐ Mixture — FULL RICH
- ☐ Carb Heat — ON
- ☐ Throttle — OPEN
- ☐ Power restored → Resume normal operations

ENGINE RESTART — IN FLIGHT (POH)

- ☐ Fuel Selector — FULLEST
- ☐ Fuel Pump — ON
- ☐ Mixture — FULL RICH
- ☐ Carb Heat — ON
- ☐ Throttle — 1" OPEN
- ☐ Magnetos — BOTH
- ☐ Primer — AS REQUIRED
- ☐ Engine Gauges — CHECK

FORCED LANDING

- ☐ Airspeed — **100 MPH**
- ☐ Landing Site — SELECT
- ☐ Mixture — IDLE CUTOFF
- ☐ Fuel Selector — OFF
- ☐ Fuel Pump — OFF
- ☐ Magnetos — OFF
- ☐ Master — OFF
- ☐ Seatbelts — SECURE
- ☐ Door — UNLATCH
- ☐ Flaps — AS REQUIRED

ENGINE FIRE — IN FLIGHT

- ☐ Radio call — MAYDAY, MAYDAY, MAYDAY on 121.5
- ☐ Squawk — 7700
- ☐ Mixture — IDLE CUTOFF
- ☐ Fuel Selector — OFF
- ☐ Fuel Pump — OFF
- ☐ Magnetos — OFF
- ☐ Cabin Heat / Air — OFF
- ☐ Airspeed **125 MPH (TOP WHITE ARC)** → Proceed to **Engine Failure / Forced Landing**

ELECTRICAL FIRE / SMOKE (POH-ALIGNED)

- ☐ Radio call — MAYDAY or PAN-PAN on 121.5
- ☐ Squawk — 7600 if comms fail; otherwise 7700
- ☐ Master Switch — OFF
- ☐ All Electrical Switches — OFF
- ☐ Cabin Heat — OFF
- ☐ Air / Vents — OPEN (ventilate)
- ☐ Smoke dissipating

- ☐ Master — ON
- ☐ Essential Circuits **ONE AT A TIME**
- ☐ If smoke returns → Master OFF → **Land ASAP**

ALTERNATOR FAILURE / NO CHARGE

- ☐ Radio call — advise ATC / request assistance and vectors
- ☐ Squawk — as assigned; 7600 if radios fail
- ☐ Electrical Load — REDUCE
- ☐ Alternator Field CB — CHECK / RESET
- ☐ Alternator Output CB — CHECK / RESET
- ☐ Alternator — OFF (30 sec) → ON
- ☐ Ammeter shows system load
- ☐ If no output → Alternator OFF → Minimal load → **Land ASAP / Practical**

IFR ACRONYMS / MNEMONICS

- ☐ IFR - Instrument Flight Rules
- ☐ VFR - Visual Flight Rules
- ☐ CDI - Course Deviation Indicator
- ☐ DME - Distance Measuring Equipment